Social Media Engagement Analytics Report

1. Objective

This report explains what was found when Python was used to study social media engagement. The goal was to understand which posts get the most likes, comments and shares, and which users and devices perform best.

2. About the Data

The dataset has 5,000 posts and 19 columns, such as age, gender, country, post type, category, likes, comments, shares, impressions, watch time, device, sentiment and followers. The posts were made between January 2022 and December 2023.

3. What Was Done

- **Loading:** The CSV file was loaded with Pandas, and the date column was changed to a proper date type.
- **Cleaning:** 150 values were missing in each of age, gender, likes, comments, shares and sentiment. Numbers were filled with the median and text with the most common value (mode). No duplicate rows were found.
- **Removing wrong values:** 600 posts had more likes, comments and shares than impressions, which is not possible, so they were removed. 4,400 posts remained.
- **New columns:** Hashtag count, engagement score (likes + comments + shares) and log of likes were created.
- **Analysis:** Averages, medians, standard deviation, percentiles and group summaries were calculated, and 15 charts were made using Matplotlib, Seaborn and Plotly.

4. Key Findings

Content performance

- Video posts had the highest average engagement score (12,288), but image, text and reel posts were very close, within about 1%.
- Food was the best category (12,431) and fitness was the lowest (12,003).
- Germany had the highest average engagement rate (0.310) and France the lowest (0.267).

User trends

- Engagement goes down a little as age goes up. The 51-64 group scored lowest, but the link between age and engagement is very weak (correlation -0.03).
- Verified accounts did not do better. Their engagement score was 11,830, compared with 12,280 for other accounts.

Behaviour

- The data has only dates, not clock times, so the best time of day could not be found. Saturday had the most impressions on average (56,802) and October was the best month (57,419).
- Mobile users watched the longest on average (4,078 seconds), ahead of tablet (3,964) and desktop (3,929).

Sentiment

- Positive posts had the highest engagement score (12,334) and neutral posts the lowest (12,041). Negative posts did almost as well as positive ones (12,291), but there were fewer of them (830 out of 4,400).

5. Things to Keep in Mind

The differences between groups are small in almost every comparison, so the patterns are weak. Removing 600 posts may also have changed some averages slightly.

6. Conclusion

Video posts, food content, mobile users and positive sentiment performed slightly better than the others, but no single factor strongly decides engagement in this data. A bigger dataset that also includes the time of posting would give clearer answers.